Exact Total Domination Numbers of Cylindrical Grids of Widths Five, Six, and Seven

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Abstract

We determine the total domination number of the cylindrical grids $P_m \square C_n$ for every circumference $n \geq 3$ when $m \in \{5, 6, 7\}$. The proof represents a total dominating set by a closed walk in a finite column automaton whose states record selected rows and the rows still requiring domination from the next column. The graph parameter is thereby the minimum diagonal entry of a min-plus matrix power. For widths five, six, and seven we verify exact entrywise matrix identities with periods four, fourteen, and four, respectively. Exact finite certificates supplied with the paper establish these identities and the finite prefix values needed to complete the computer-assisted part of the proof. The resulting formulas include the exceptional circumferences $n = 12$ for width six and $n = 7, 14$ for width seven.

Keywords: total domination; cylindrical grid; Cartesian product; finite-state method; min-plus algebra; computer-assisted proof.

1 Introduction

Let G be a graph without isolated vertices. A set $D \subseteq V(G)$ is *totally dominating* if every vertex of G, including every vertex of D, has a neighbor in D. The minimum cardinality of such a set is the total domination number $\gamma_t(G)$. In this paper we study the cylindrical grids $P_m \square C_n$, the Cartesian product of a path and a cycle.

Total domination in grid-like products has a substantial literature. Gravier studied total domination in products of paths and cycles and related tiling formulations [1]. Exact cylindrical results for smaller fixed path widths were obtained by Eakawinrujee for widths two, three, and four [3]; the author's dissertation gives additional provenance for that cylinder program [4]. Rectangular grids, ordinary domination, and several variants provide nearby but distinct problems; for example, the fixed-width results of Crevals and Ostergard concern $P_m \square P_n$, not cylinders [2], while Nandi, Parui, and Adhikari study ordinary domination [6].

The results below concern widths not covered by the fixed-width cylindrical formulas cited above. Hu, Sohn, and Chen study related bundle and toroidal cases, including C_m bundles over C_n and an exact result for $C_n \square C_5$; these are important context but are not silently identified with the cylinder $P_5 \square C_n$ [5]. Eakawinrujee's public dissertation records the exact subfamily

$$p \equiv 1 \pmod 2, \quad q \equiv 0 \pmod 4 \quad \Longrightarrow \quad \gamma_t(P_p \square C_q) = \frac{(p+1)q}{4},$$

which overlaps the present width-five and width-seven formulas on infinitely many circumferences [4]. We therefore make the narrow claim of complete classifications for widths five, six, and seven across all $n \geq 3$ in the audited record, extending rather than erasing these earlier exact subfamilies. The cyclic boundary condition couples the first and last columns, and a selected vertex cannot dominate itself. Our approach handles both issues with a finite-state representation; the semantic reduction is proved in the paper and the finite identities are supplied as exact computer-assisted certificates.

The methodological distinction from recent nearby work is also important. Garzón, Martínez, Moreno, and Puertas already use min-plus algorithms for 2-domination of small-cycle cylinders [8]. Martínez, Castaño-Fernández, and Puertas later use tropical matrix products for broader 2-domination cylinder bounds [7]; that invariant requires two selected neighbors for vertices outside the set, whereas ordinary total domination requires one open neighbor for every vertex, including selected vertices. The shared min-plus language is methodological context, not an equivalence of parameters.

1.1 Prior results and scope of the contribution

The relevant boundary of the literature can be summarized as follows. The first two rows concern the same total-domination parameter, while the latter two provide context or a nearby invariant.

source	scope relevant to this paper
Eakawinrujee (2022)	exact cylinder results for smaller fixed path widths; bounds and related results beyond that range
Eakawinrujee dissertation	exact odd path-width subfamily $q \equiv 0 \pmod 4$, including infinitely many width-five and width-seven cases
Hu–Sohn–Chen (2016)	related bundle/toroidal total and paired domination results, including an exact $C_n \Box C_5$ case
Garzón–Martínez–Moreno–Puertas (2021)	min-plus algorithms for 2-domination of small-cycle cylinders, a different invariant
Martínez–Castaño-Fernández–Puertas (2024)	finite-state/tropical methods for 2-domination of cylinders, a different invariant

The present result is not a claim that any one of these ingredients is new in isolation. Its bounded claim is that the exact values for every $n \geq 3$ are supplied for each of the three widths $m = 5, 6, 7$, with the earlier exact subfamilies and related parameters explicitly separated from the all-circumference statement.

Our main results are the following.

Theorem 1.1 (Width five)**.** For every $n \geq 3$,

$$\gamma_t(P_5 \,\Box\, C_n) = \begin{cases} \lceil 3n/2 \rceil + 1, & n \equiv 2 \pmod 4, \\ \lceil 3n/2 \rceil, & \text{otherwise.} \end{cases}$$

Theorem 1.2 (Width six)**.** Let ε_r, for $r \in \{0, \dots, 13\}$, be given by

$$\varepsilon_r = \begin{cases} 0, & r \in \{0, 3, 5, 7, 9, 10\}, \\ 1, & r \in \{1, 4, 6, 11, 13\}, \\ 2, & r \in \{2, 8\}, \\ 3, & r = 12. \end{cases}$$

Then

$$\gamma_t(P_6 \,\Box\, C_n) = \left\lceil \frac{12n}{7} \right\rceil + \varepsilon_{n \bmod 14}$$

for every $n \geq 3$, except that $\gamma_t(P_6 \,\Box\, C_{12}) = 22$.

Theorem 1.3 (Width seven)**.** For every $n \geq 3$,

$$\gamma_t(P_7 \,\Box\, C_n) = 2n,$$

except $\gamma_t(P_7 \,\Box\, C_7) = 15$ and $\gamma_t(P_7 \,\Box\, C_{14}) = 30$.

The proof has two layers. First, we prove the graph-theoretic equivalence between total dominating sets and weighted closed walks. Second, a finite certificate establishes the matrix identities and prefix values. The thresholded recurrence lemma then shows exactly how those finite facts imply each all-circumference formula.

For orientation, exact small-circumference cylinder cases are already part of the prior record discussed by Hu–Sohn–Chen and reproduced in the public Eakawinrujee dissertation. For the three target widths, those prior C_3- and C_4-values are $5, 6, 6, 6, 8, 6, 8$, respectively. The same values are independently present in our finite prefix, but are not claimed as new. The table separates these checked small circumferences from the all-circumference classification supplied below; it does not attribute the cylinder cases to Hu–Sohn–Chen's separate toroidal $C_n \square C_5$ result:

m	$\gamma_t(P_m \,\square\, C_3)$	$\gamma_t(P_m \,\square\, C_4)$
5	5	6
6	6	8
7	6	8

2 Preliminaries

We use $[m] = \{0, 1, \ldots, m-1\}$ for the rows of P_m, and index the cycle columns by $\mathbb{Z}/n\mathbb{Z}$. For $S \subseteq [m]$, define its open path neighborhood by

$$N_P(S) = \{i \in [m] : i-1 \in S \text{ or } i+1 \in S\},$$

where indices outside $[m]$ are omitted. Thus a row selected in a column does not automatically dominate the vertex in that same row and column.

For a set $D \subseteq V(P_m \,\square\, C_n)$, write

$$S_j = \{i \in [m] : (i, j) \in D\}.$$

The horizontal neighbors of (i, j) are $(i, j-1)$ and $(i, j+1)$, and the vertical neighbors are the vertices in rows adjacent to i in the same column.

3 The finite-state model

For a cyclic sequence of selected row sets (S_j), define the pending set

$$R_j = [m] \setminus \bigl(S_{j-1} \cup N_P(S_j)\bigr).$$

The rows in R_j are precisely those for which the left horizontal and vertical domination alternatives are absent in column j. Define the fixed admissible state set

$$Q_m = \{(S, R) : S \subseteq [m],\ R \subseteq [m] \setminus N_P(S), \text{ and } R = [m] \setminus (U \cup N_P(S)) \text{ for some } U \subseteq [m]\}.$$

Thus the predecessor mask is quantified in the definition but is not part of the state. Given a state $q = (S, R) \in Q_m$ and an input mask $T \subseteq [m]$, the transition is legal precisely when

$$R \subseteq T,$$

and its head pending set is forced to be

$$R' = [m] \setminus \bigl(S \cup N_P(T)\bigr).$$

The edge is $(S, R) \to (T, R')$, where the displayed formula defines R', provided $R \subseteq T$ and $(T, R') \in Q_m$; its weight is $|T|$. Every edge head is therefore a member of the fixed finite set Q_m.

Theorem 3.1 (Finite-state representation)**.** For $m \in \{5,6,7\}$ and $n \geq 3$, labeled total dominating sets of $P_m \,\square\, C_n$ are in weight-preserving bijection with labeled length-n closed walks in the state graph above. The weight of a walk is the sum of its transition weights.

Proof. We first prove soundness. Let a cyclic legal walk be given and define $D = \{(i,j) : i \in S_j\}$. Consider a vertex (i,j). If $i \notin R_j$, then by the definition of R_j, either $i \in S_{j-1}$ or $i \in N_P(S_j)$. Hence (i,j) has a selected left or vertical open neighbor. If $i \in R_j$, transition legality gives $i \in S_{j+1}$, so the right horizontal neighbor is selected. These cases include selected vertices themselves and therefore establish total domination.

Conversely, let D be a total dominating set and define S_j from its selected vertices and R_j by the displayed formula. If $i \in R_j$, then the left and every vertical selected-neighbor alternative is absent. Total domination forces the right neighbor $(i, j+1)$ to be selected, so $R_j \subseteq S_{j+1}$. The next pending formula follows from its definition. The same argument applies to the wraparound transition.

The two constructions are inverse because the pending masks are forced by the selected masks. Finally,

$$\sum_j |S_{j+1}| = \sum_j |S_j| = |D|,$$

so weights agree. For $n = 3$, the left and right horizontal neighbors are distinct because $3 \nmid 2$, so the cyclic encoding has exactly the simple graph neighborhood semantics. $\square$

3.1 A local transition example

It is useful to see the pending-mask convention in a concrete transition. Take $S = \varnothing$ and $R = [m]$. This is an admissible state: before the current column is read, every row may still require its right horizontal neighbor. Choose $T = [m]$. The edge is legal because $R \subseteq T$, and its head pending set is

$$[m] \setminus \big(\varnothing \cup N_P([m])\big) = \varnothing.$$

The transition has weight m. In the corresponding two-column fragment, the selected vertices in T are dominated vertically, while the rows that were pending in the tail receive their required right neighbor. This example also illustrates why the state records obligations created by the preceding column rather than treating a selected vertex as self-dominating.

More generally, the state condition is not an additional graph restriction. For $R \subseteq [m] \setminus N_P(S)$, taking

$$U = [m] \setminus \big(R \cup N_P(S)\big)$$

gives $R = [m] \setminus (U \cup N_P(S))$. Thus the quantified predecessor mask in the definition of Q_m is exactly the realizability condition for the residual obligation mask. The independent state reconstruction uses this equivalent characterization.

4 Min-plus path semantics and the recurrence certificate

Let M_m be the weighted adjacency matrix of this graph over $\mathbb{Z}_{\geq 0} \cup \{+\infty\}$, with $+\infty$ for a missing edge. Min-plus addition and multiplication are

$$a \oplus b = \min(a,b), \qquad a \otimes b = a + b,$$

with $+\infty$ absorbing for $\otimes$. Matrix powers use min-plus multiplication. The min-plus identity matrix has zero diagonal and $+\infty$ off the diagonal. For $c \in \mathbb{Z}_{\geq 0}$, scalar matrix shift means $(c \otimes A)_{uv} = c + A_{uv}$ for finite A_{uv}, and $(c \otimes A)_{uv} = +\infty$ otherwise.

Lemma 4.1 (Min-plus path semantics)**.** For every $k \geq 0$, $(M_m^k)_{uv}$ is the minimum weight of a length-k walk from u to v, with the minimum of an empty set interpreted as $+\infty$.

Proof. The case $k = 0$ is the min-plus identity matrix. For the induction step, partition a length-$k+1$ walk by its penultimate state x. The prefix has minimum cost $(M_m^k)_{ux}$, the final edge has cost $(M_m)_{xv}$, and minimizing over the finite state set gives the min-plus product. Missing prefixes or edges contribute $+\infty$ and therefore do not create a walk. $\square$

Define

$$g_m(n) = \min_{q \in Q_m} (M_m^n)_{qq}.$$

Theorem 3.1 and Lemma 4.1 give

$$\gamma_t(P_m \,\square\, C_n) = g_m(n), \qquad n \geq 3.$$

The distinction between an entrywise identity and a scalar sequence pattern is important. An identity for the minimum diagonal alone would not control arbitrary endpoints and would not justify concatenating an additional walk. The certificates instead compare every matrix entry at the two exponents. This is the reason the recurrence lemma applies uniformly to all closed walks, including those whose minimizing state changes with the circumference.

Lemma 4.2 (Thresholded min-plus recurrence)**.** Suppose an entrywise identity

$$M^{N+p} = c \otimes M^N$$

holds for a finite integer c. Define $h(t) = \min_q (M^t)_{qq}$. Then for every $k \geq 0$,

$$M^{N+p+k} = c \otimes M^{N+k},$$

and hence $h(n+p) = h(n) + c$ for every finite-value base $n \geq N$.

Proof. Min-plus matrix multiplication is associative: both parenthesizations minimize the same finite collection of triple products. A finite scalar shift commutes with a subsequent min-plus product, including entries equal to $+\infty$. Therefore, for $k \geq 0$,

$$M^{N+p+k} = M^{N+p} \otimes M^k = (c \otimes M^N) \otimes M^k = c \otimes (M^N \otimes M^k) = c \otimes M^{N+k}.$$

Taking the minimum over diagonal entries gives the same statement for g. When the relevant diagonal minimum is finite, the scalar operation is ordinary addition. The conclusion is explicitly restricted to $n = N + k \geq N$; no value below the threshold is propagated by this lemma. $\square$

5 Finite certificates and exact formulas

The finite certificate proposition below is the computational input to the three theorem proofs. Its verifier reconstructs the state and transition semantics, parses the persisted matrices, recomputes both powers, and checks the entrywise identities with explicit $+\infty$ semantics. The proposition is computer-assisted: its trusted inputs are the definitions, the certificate files, and the verifier specified in the reproducibility record.

Proposition 5.1 (Finite certificate and prefix data)**.** For widths five, six, and seven, the state and transition graphs have the counts in Table 1; the displayed matrix identities hold; and the diagonal minima $g_m(n)$ agree with the exact finite-prefix tables in the project artifact on $3 \leq n \leq 19$, $3 \leq n \leq 34$, and $3 \leq n \leq 31$, respectively.

The certified identities are

$$M_5^{20} = 6 \otimes M_5^{16}, \qquad M_6^{35} = 24 \otimes M_6^{21}, \qquad M_7^{32} = 8 \otimes M_7^{28}.$$

Consequently the recurrences begin at $n = 16$, $n = 21$, and $n = 28$, respectively. The exact finite-prefix replay covers $3 \leq n \leq 19$ for width five, $3 \leq n \leq 34$ for width six, and $3 \leq n \leq 31$ for width seven. Since each interval ends at $N + p - 1$, every later index reduces to a checked base in $[N, N + p - 1]$.

The decisive finite values are displayed here. The complete machine-readable tables, including all rows from 3 through the stated prefix endpoint, are part of the reproducibility record.

Table 1: Finite state graphs and entrywise recurrence certificates.

width	states	transitions	N	p	c
5	169	2,419	16	4	6
6	441	11,025	21	14	24
7	1,156	50,303	28	4	8

Table 2: Width-five prefix values.

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$g_5(n)$	5	6	8	10	11	12	14	16	17	18	20	22	23	24	26	28	29

5.1 Width five

The identity $M_5^{20} = 6 \otimes M_5^{16}$ gives $g_5(n+4) = g_5(n) + 6$ for $n \geq 16$. *Proof of Theorem 1.1.* The finite proposition gives every value $3 \leq n \leq 19$; the base values $16 \leq n \leq 19$ are shown in Table 2, and the complete prefix comparison is in the machine-readable table. Every $n \geq 20$ reduces by four to one of these bases, and the residue $n \equiv 2 \pmod 4$ contributes the additional unit.

5.2 Width six

The identity $M_6^{35} = 24 \otimes M_6^{21}$ gives $g_6(n+14) = g_6(n)+24$ only for $n \geq 21$. *Proof of Theorem 1.2.* The value at $n = 12$ is a finite-prefix exception and is not propagated to $n = 26$. The finite proposition gives every value $3 \leq n \leq 34$, including all rows $3 \leq n < 21$ and every base value $21 \leq n \leq 34$. For every $n \geq 35$, repeated subtraction of fourteen reaches one of the checked values $21, \dots, 34$, which proves Theorem 1.2; the recurrence is never applied below its threshold.

5.3 Width seven

The identity $M_7^{32} = 8 \otimes M_7^{28}$ gives $g_7(n+4) = g_7(n) + 8$ only for $n \geq 28$. *Proof of Theorem 1.3.* The finite proposition explicitly covers every row $3 \leq n \leq 31$, including the ordinary prefix rows and the exceptions $g_7(7) = 15$ and $g_7(14) = 30$. Every $n \geq 32$ reduces to a base in $28, \dots, 31$, proving the theorem. The exceptions remain only in the finite prefix and are not propagated by the recurrence.

6 Verification and reproducibility

The repository includes a reference implementation and a separately written certificate verifier. The verifier rebuilds states, transitions, weights, and matrix powers from the definitions. Detailed test counts, mutation results, certificate manifests, and hash bindings are recorded separately from the mathematical proof. The three certificate commands are:

```
python verify/verify_certificates.py 5 certificates/width5
python verify/verify_certificates.py 6 certificates/width6
python verify/verify_certificates.py 7 certificates/width7
```

Each command reports zero identity mismatches. The repository's SHA256SUMS file binds the finite artifacts, and paper/REPRODUCIBILITY.md records the expected dimensions, counts, parameters, and verification outputs.

The computations establish the finite proposition; the preceding semantic lemmas and theorem arguments establish why that proposition implies the infinite statements.

Table 3: Width-six base and exceptional values.

n	12	21	22	23	24	25	26	27	28	29	30	31	32	33	34
$g_6(n)$	22	36	40	40	42	44	48	48	48	51	54	54	56	57	60

Table 4: Width-seven exceptional and tail-base values.

n	7	14	28	29	30	31
$g_7(n)$	15	30	56	58	60	62

6.1 What is and is not certified

The finite proposition has three logically separate inputs. First, the definition-level state and transition reconstruction fixes the finite graph. Second, the persisted matrices are parsed with explicit finite and $+\infty$ tags, and the verifier recomputes both powers from the reconstructed edges rather than trusting a producer's claimed identity. Third, the prefix trace initializes every state as a possible source and minimizes the matching diagonal entry, so its values are the unrestricted $g_m(n)$, not a low-weight-source surrogate. Hashes bind the persisted binary artifacts to the manifests used by the verifier.

This separation also fixes the boundary of the computer-assisted argument. The code supplies finite exact premises; it does not replace the total-domination bijection, min-plus path lemma, or thresholded algebraic implication proved above. Conversely, the written argument does not assert that a finite set of tested values alone proves an infinite formula: the entrywise identities and their thresholds are the step that supplies the tail.

7 Discussion

The three widths have different eventual periods: four for widths five and seven, and fourteen for width six. This difference is visible in the finite state graphs and in the residue correction for width six. The method scales naturally as a fixed-width transfer construction, but the finite state space grows rapidly with the width. No claim about width eight is made here.

The width-five and width-seven formulas have the same eventual increment pattern but arise from different state spaces, while width six has a longer period and a nontrivial residue correction. The period-four tails are visible in the corresponding entrywise identities, whereas the period fourteen in width six is an exact property of the certified matrix power, not an assumed periodicity ansatz. The exceptional values are consequently treated as finite-prefix data: they are not extrapolated from a recurrence whose base threshold excludes them.

The contribution should be read as a fixed-width frontier result and a reusable verification pattern. Earlier work settles smaller widths or particular residue subfamilies, and nearby 2-domination work demonstrates that tropical transfer methods are useful for cylindrical problems. The present package adds complete all-circumference total-domination formulas for the three frozen widths, together with a clean-room certificate protocol that separates graph semantics, finite matrix identities, and tail induction. No claim is made that the transfer architecture itself is new in general, or that the method resolves arbitrary width.

A cyclic domination problem can be expressed as a closed-walk problem without sacrificing the open-neighborhood condition. Once the finite weighted graph is fixed, an entrywise min-plus identity controls every subsequent walk length at and above its threshold; the finite prefix accounts for the transients and exceptional circumferences.

8 Conclusion

We determined the exact total domination numbers of $P_5 \,\square\, C_n$, $P_6 \,\square\, C_n$, and $P_7 \,\square\, C_n$ for every $n \geq 3$. The proof combines a weight-preserving finite-state bijection, min-plus path semantics, three exact entrywise matrix identities, and finite-prefix verification. The computer-assisted portion is specified by the finite proposition and the reproducibility record, while the graph-theoretic reduction and recurrence argument are given explicitly above.

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